SANDBOX · PROFILE STUDIO — TUTORIAL 3

The process stack

After the material stack and opening are built, the process stack transforms the profile step by step — the same idea as real fab steps (deposit, fill, etch, polish).

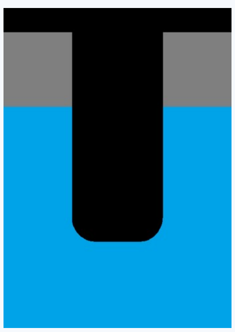
How it works

Steps run **in order, top to bottom** (①, then ②, ③ …). Each step acts on the result of the one above it. A **Repeat** block runs its steps ×N — handy for multilayers / superlattices.

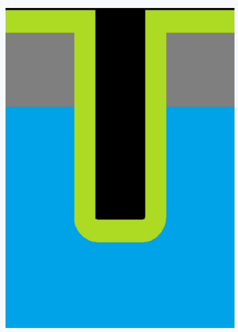
The step types

- **Deposit · conformal** — coats every surface (walls, bottom, top) with a uniform layer that follows the profile. Use it for liners and spacers.
- **Deposit · planar** — adds a flat layer on top only (no sidewall coverage).
- **Fill** — gap-fills the opening flush with the surface. The number is **overfill** (extra nm of blanket above the surface; 0 = flush).
- **Etch · isotropic** — removes material equally in all directions; rounds corners and undercuts.
- **Etch · anisotropic** — removes material vertically. Anisotropy 1 = fully vertical, 0 = isotropic, in between = a rounded undercut. Pick a material to etch only that one, or “(any)”.
- **Planarize** — cuts everything flat at a height, like chemical-mechanical polish (CMP).

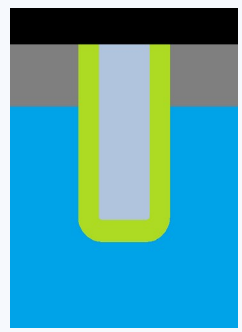
Worked example — liner + fill + polish



① bare trench (stack + opening)



② + Deposit·conformal (nitride liner)



③ + Fill (tungsten) + Planarize

Tips

- Deposit / Fill can use ANY material in the palette — you don’t need to add liner/fill materials to the stack first. Etch lists the materials present (plus “(any)”).
- Etch with a material selected removes only that material; “(any)” removes whatever it reaches.
- Build a superlattice with a Repeat block: e.g. deposit A, deposit B, repeated ×10.